AP CALCULUS AB

Unit 1 • Self Check 1.1 • Lessons 1-1 and 1-2

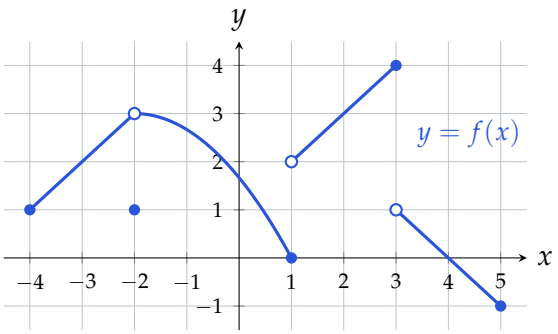
Self Check 1.1 • Lessons 1-1 and 1-2

Name: _____ Date: _____

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Questions 1 and 2 refer to the graph of f.



Question 1

NO CALCULATOR

Using the graph of f, $\lim_{x \to -3} f(x) =$

(A) 1
(B) 2
(C) 3
(D) It does not exist.

Question 2

NO CALCULATOR

Using the graph of f, $\lim_{x \to 3^-} f(x) =$

(A) 1
(B) 3
(C) It does not exist.
(D) 4

Question 3

NO CALCULATOR

$\lim_{x\to 1} \dfrac{x^2+2x-3}{x-1} =$

(A) 4
(B) 0
(C) 2
(D) It does not exist.

Question 4

NO CALCULATOR

Let $j(x) = \begin{cases} 2x+3, & x < -1 \\ x^2, & x \geq -1. \end{cases}$ Then $\lim_{x\to -1} j(x) =$

(A) -1
(B) It does not exist.
(C) 1
(D) 3

Question 5 (Free response)

NO CALCULATOR

Find $\lim_{x\to 0} \dfrac{x^2+5x}{x}$, $\lim_{x\to -3} \dfrac{x^2-9}{x+3}$, and $\lim_{x\to 2} j(x)$ for $j(x) = \begin{cases} x^2, & x < 2 \\ 6-x, & x \geq 2. \end{cases}$